

10.6 A Hybrid-Bonded 12.1TOPS/mm² 56-Core DNN Processor with 2.5Tb/s/mm² 3D Network on Chip

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Abstract

A **manycore** DNN processor leverages hybrid bonding in a **14×4×2 mesh network** on chip to increase 3D bandwidth between cores, accelerators, and SRAM to **2.5Tb/s/mm²**. Memory throughput is improved by 39% compared to an equivalent 2D design with the same

memory capacity. The **56 16×16 INT8 systolic array** accelerators are stacked on top of 56 RISC-V cores to enable a peak AI performance density of 12.1TOPS/mm². The DNN processor scores a latency of 53μs on the image classification benchmark from MLPerf.

AI and machine-learning workloads feature dense data movement and high-bandwidth communication between cores, accelerators, and memory. Reducing the power and performance overhead of this data movement is key for energy-efficient operation. **As die area increases, cross-die communication can become a critical bottleneck, limiting workload performance.** 3D stacking addresses this communication challenge, but achieving the full throughput benefits of 3D requires high-bandwidth connectivity between dies, such as that enabled by **hybrid-bonded interconnect (HBI)**. In manycore processors, HBI enables both wide memory access from parallel processing cores and lower latency to shared distributed memory. 3D networks on chip (NoCs) provide low-latency access to memory on both top and bottom dies, **with cores agnostic to 3D memory design.** Further, **creating a general-purpose manycore bottom die allows design extensibility,** with the base die used either stand-alone or augmented with application-specific top dies **matching a common 3D NoC and HBI footprint.** For DNN applications, multiply accumulate (MAC) operations can be offloaded to accelerators in the top die, while the bottom die orchestrates 3D data movement.

A manycore 3D DNN processor (M3DProc) is implemented with **9μm HBI** connecting **56 RISC-V core tiles** in an **Intel 3 bottom die** to **56 DNN accelerator tiles** in a **12.1TOPS/mm² Intel 18A top die** (Fig. 10.6.1). Cores and DNN accelerators access data from a shared **5.25MB SRAM** that is distributed across the stacked dies. A **14×4×2 mesh 3D NoC (M3DNoC)** delivers memory accesses with 2.5Tb/s/mm² throughput across the HBI. M3DNoC improves memory throughput by up to 39% compared to 2D NoCs spanning an identical memory capacity. **M3DProc has a peak dynamic energy efficiency of 16.1TOPS/W excluding leakage at 0.5V, 280MHz.** The chip scores a latency of 53μs on the image classification benchmark from MLPerf Tiny.

Manycore processors are designed to maximize compute density for increased core count and greater on-chip physical address space, often eliminating features such as out-of-order operation and coherent shared memory [1][2]. In M3DProc, the 56 bottom tiles each contain a 32b RISC-V (RV32) core, modified from open-source RTL [3]. **The cores use the 5.25MB on-chip memory as a partitioned global address space,** giving all cores access to memory in all top and bottom tiles. The 4-entry fully associative data (D\$) and instruction (I\$) caches **occupy only 0.5% of the bottom tile area to maximize total shared memory.** **Write-through store operations to 64b cache lines** eliminate modified data tracking and simplify memory control.

M3DNoC provides low-latency data transfer between compute elements and memory, while also **allowing for seamless design extensibility** (Fig. 10.6.2). M3DProc size can be increased via simple abutment of tiles on both the top and bottom dies, which share the same area footprint. M3DNoC has a 14×4×2 3D mesh topology which spans both top and bottom dies as well as connecting off-chip through the chip bridge. It performs **dimension-order routing with two separate physical channels for requests and responses.** M3DNoC uses wormhole flow control with header packets containing addresses followed by one or more 64b data flits. Routers use valid/ready signaling with single-cycle latency to reduce in-network register requirements. Input and output port latches work in tandem to support full-throughput operation [4], while requiring 50% fewer sequential elements compared to a 2-entry FIFO. **The data latches are clock gated (CG)** by the valid and ready signals to reduce active idle power.

3D NoCs must handle communication between layers in the 3D design, which is challenging since **uncorrelated process variation on the two dies induces clock skew that impacts timing margin** [5]. Balancing clock paths with post-silicon de-skewing and reducing the length of divergent clock paths with numerous 3D clock connections [6] have been proposed to address these issues. To simplify design construction, increase modularity, and facilitate timing closure, the M3DNoC HBI interface uses a **forwarded-clock design, which alleviates timing constraints by allowing clock delay to scale together with data between dies.** While this incurs a latency penalty when crossing between dies, DNN workloads are more throughput oriented, making this a good tradeoff for power, performance, and ease of design. **A credit-based clock-crossing circuit synchronizes data to the local clock as it crosses the 3D interface,** using request and acknowledge HBI pads accompanying the data in each direction, with an 8-entry asynchronous FIFO enabling full bandwidth across the 3D interface.

The top die pairs with the RISC-V base die and M3DNoC to offer application-specific acceleration. In this design, each of the 56 top tiles contains a DNN accelerator that implements convolution (Conv) **with bit accuracy to a Tensorflow Lite software baseline** (Fig. 10.6.3). Computation consists of matrix-matrix multiplication performed with a **16×16 INT8 systolic array, followed by bias addition, batch normalization, and activation function units.** Conv is divided into operations for 4×4 height (h) by width (w) output feature map patches with 16 output channels (k) and is either distributed spatially or processed serially by accelerators. **Cores initiate DNN computation through memory-mapped communication, and accelerators signal completion via an interrupt (IRQ), enabling concurrent RISC-V and accelerator processing.** The memory-mapped registers also hold Conv scalar input parameters such as tensor pointers and layer dimensions. Input (X), output (Y) data, and weights (W) are stored channel-first, with INT8 values packed into 64b words. Per-channel parameters for element-wise operations are also read from memory. The accelerator transposes and lowers input activations before systolic array computations, and the output register file has capacity for two output feature map patches for double buffering. The accelerator latency is $s \times r \times c + 34$ cycles, where s and r are the filter dimensions, and c is the number of input channels.

For high-throughput data movement, each of the 112 tiles contains a direct memory access (DMA) engine that supports memory copy operations for up to 2kB of contiguous memory at a time. DMA is used for Conv operations across multiple accelerators with the same input feature map patch. Burst DMA transfers are transported over the NoC as a single packet, with one header flit followed by up to 256 data flits. Similar to the DNN accelerator, the DMA is configured via memory-mapped registers and returns an interrupt to the core when complete. Core-to-core interrupts are also used to create hardware threads and perform barrier synchronization. Any core can create a thread by sending a function pointer, input parameter pointer, and interrupt to an idle follower core. The follower core's interrupt handler executes the function and returns an interrupt to the leader core. Leader cores perform a 'join' operation to wait for thread completion and complete a barrier synchronization by performing 'join' for all active threads.

M3DProc is implemented with **an Intel 3 [7] bottom die** and an **Intel 18A [8] top die,** connected by **9μm HBI** (Fig. 10.6.7). It is part of a larger test vehicle with a shared power rail, so dynamic energy is reported by subtracting out the measured full-chip leakage energy at each data point. M3DProc is measured across a wide range of voltages and frequencies (Fig. 10.6.4) **for a peak TOPS workload, achieving a peak performance of 33.0TOPS at 1.1V, 1.205 GHz, where one MAC is 2 OPs.** The chip was measured down to 0.5V where the **peak dynamic energy efficiency is 16.1TOPS/W at 280MHz.** M3DNoC throughput is measured for 2D and 3D random uniform access patterns across a 2.6MB address space, where traffic is generated by RISC-V cores with varying transfer sizes. The 2D traffic pattern accesses all 56 bottom tiles, while the 3D test accesses 28 tiles in each of the top and bottom dies. Larger transfer sizes amortize the overheads of core operation; however, **2D traffic saturates at a lower transfer size due to NoC congestion.** 3D communication benefits from higher bisection bandwidth and improves throughput by up to 39%. Energy for 2D and 3D DMA memory copy operations is measured across supply voltages for nearest-neighbor communication between all 112 tiles. For 2D operation, all tiles communicate with their neighboring tile on the same die, while for 3D operation, each tile communicates with the tile stacked directly above/below across the HBI interface. The DMA energy measurement has multiple components including core DMA initialization, NoC transfer, and core interrupt handling of the DMA completion interrupt. Measurements show no energy penalty for 3D DMA transfers.

Performance is measured for the image classification benchmark from MLPerf Tiny, which performs inference for CIFAR-10 using ResNet-8 [9]. Since each processor core runs an independent thread, the design enables flexible workload mappings that are optimized for different metrics such as latency and energy. Two alternative mappings are written in C++, compiled with GCC, and run with weights in on-chip memory (Fig. 10.6.5). A parallel mapping optimizes for the best end-to-end latency by parallelizing each DNN layer across as many cores as possible. It processes layers sequentially with barrier synchronization between layers to resolve core-to-core dependencies. Independent layers are computed in parallel, such as Conv layers 3 and 5, and 'Add' operations are fused with Conv layers 2

and 4. 'Add' and 'Pool' operations contribute significantly to the inference latency since they contain many calculations performed in the RISC-V cores. The parallel mapping scores a latency of 53 μ s on the MLPerf image classification benchmark [10]. An alternative dataflow workload mapping optimizes for lowest inference energy by reducing cross-chip communication, trading off end-to-end latency. This dataflow mapping allocates a subset of cores for each DNN layer, passing results between layers in a pipelined fashion [11]. Conv layer 7 and the fully connected layer are processed in the same core since neither of these operations limit overall throughput. By confining layer computation to a smaller portion of the chip communication overheads are reduced, improving throughput by 1.5 $\times$ to 27.4k inferences/sec at 1.1V. Energy is also reduced by 1.5 $\times$ to as low as 15.6 μ J/inference at 0.5V.

Figure 10.6.6 compares M3DProc to previous works on HBI interfaces and manycore DNN processors. M3DNoC employs higher HBI signal density than previous works, enabling a 3D throughput density of up to 2.5Tb/s/mm². The combination of 3D stacking and process scaling enables M3DProc to achieve higher compute density than previous manycore DNN processors.

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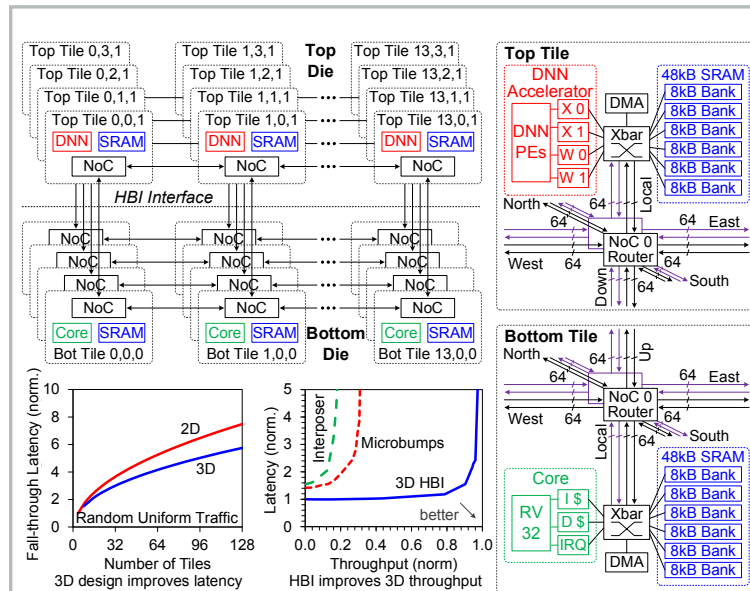


Figure 10.6.1: Block diagram of manycore 3D DNN processor and motivation for HBI.

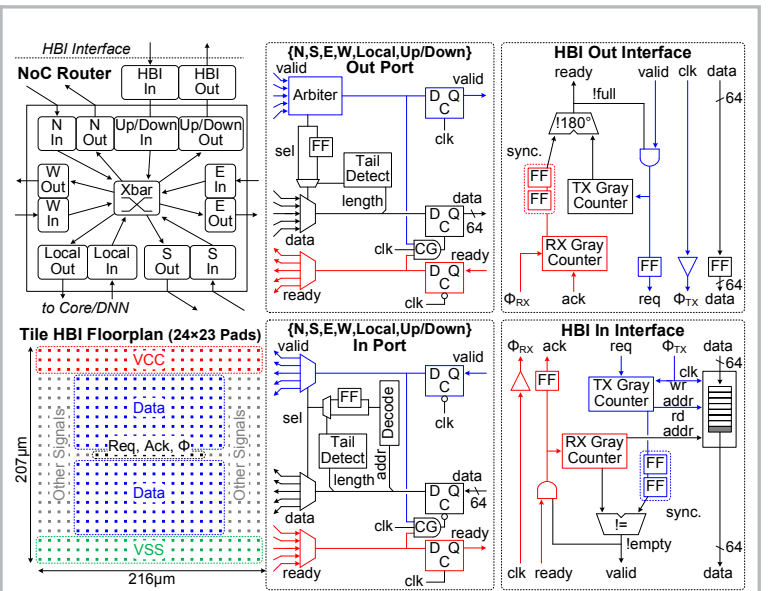


Figure 10.6.2: Mesh 3D network on chip and hybrid-bonded interface.

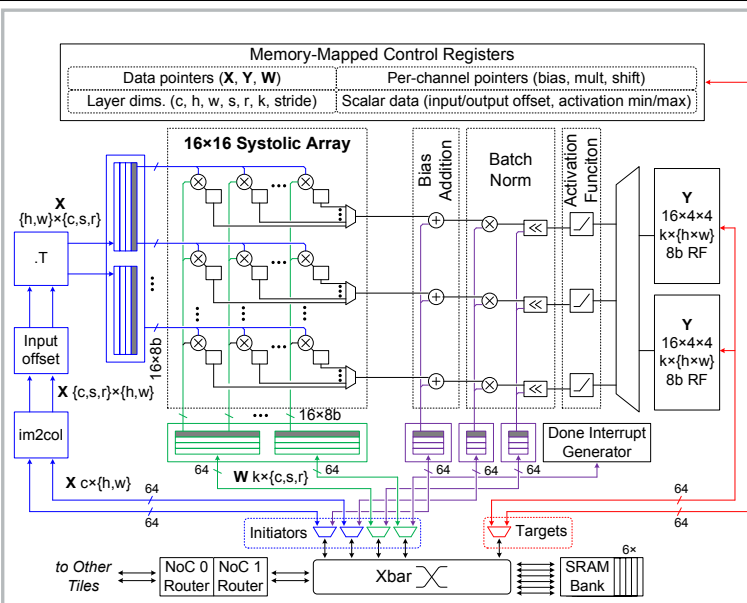


Figure 10.6.3: Block diagram of DNN accelerator in the top tile.

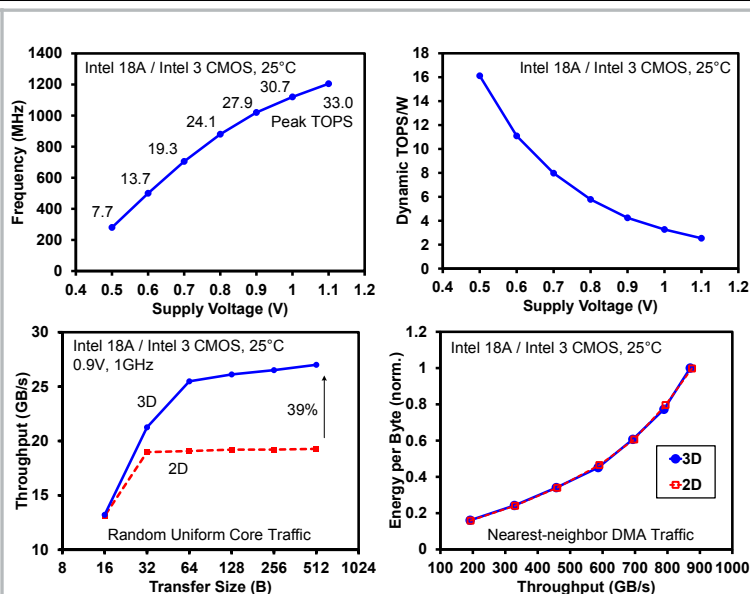


Figure 10.6.4: Chip measurements and comparison of 2D vs. 3D traffic.

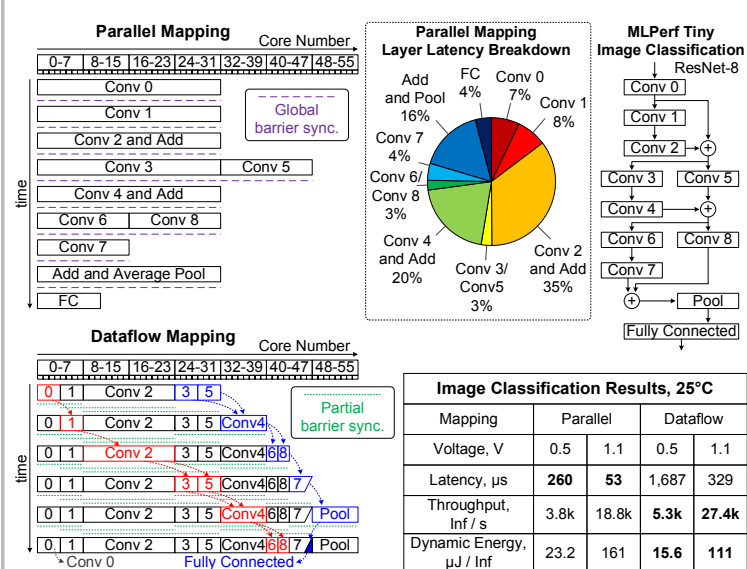


Figure 10.6.5: Image classification measurements for parallel and dataflow mappings.

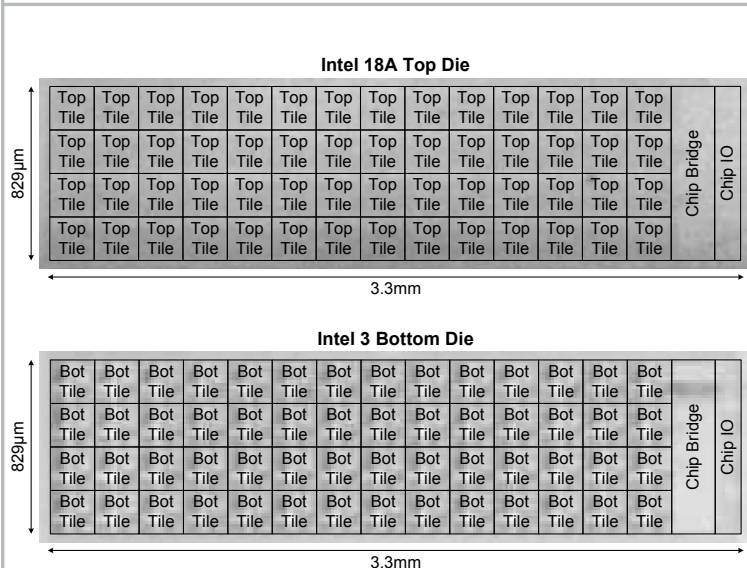


Figure 10.6.7: Top and bottom die photos of 3D stacked DNN processor.

Hybrid Bonded Chips	ISSCC22 [12]	ISSCC22 [13]	ISSCC24 [6]	ISSCC24 [14]	This Work
Top Process	7nm	25nm	7nm	5nm	Intel 18A
Bottom Process	7nm	55nm	7nm	6nm	Intel 3
Bonding Technology	HBI 9 μ m	HBI 3 μ m	HBI <2 μ m	HBI 9 μ m	HBI 9 μ m
Area, mm ²	81	602.22	15.17	4 $\times$ 1/2 Reticle	2.74
Voltage, V	-	1.1, 1.2	0.58-0.69	-	0.5-1.1
Frequency, GHz	-	0.15, 0.3	0.4-0.6	2.1	0.280-1.205
3D Signal Count	-	-	33.7 k	1.4 M*	21.5 k
Peak 3D Bandwidth, Tb/s	16	11.04	6.6 [†]	-	7.0
3D Signal Density, / mm ²	-	-	2.2 k	-	7.9 k
Peak BW Density, Tb/s/mm ²	0.020	0.018	0.44	-	2.5

*Includes both signal and power hybrid bonded connections. [†]6.1 Tb/s local memory + 512 Gb/s SMEM

DNN Processors	ISSCC24 [14]	ISSCC24 [15]	ISSCC24 [16]	ISSCC24 [17]	ISSCC25 [11]	ISSCC25 [18]	This Work
Process	5nm, 6nm	5nm	12nm	12nm	5nm	5nm	Intel 18A, Intel 3
Area, mm ²	4 $\times$ 1/2 Reticle	149	144	795	2 $\times$ 600	653	2.74
Voltage, V	-	-	0.68	0.8	-	0.75	0.5-1.1
Frequency, GHz	2.1	-	0.800	0.4-0.5	-	1.0	0.280-1.205
SRAM Capacity, MB	256*	64 [†]	52 [‡]	224 [§]	520	256	5.25
Data Precision	BF16	INT8	INT8	INT8	BF16	INT8	INT8
Peak Perf., TOPS/TFLOPS	1307	128	209.6	200	640	512	33.0
Comp. Density, TOPS/mm ²	-	0.86	1.46	0.25	2.13 [‡]	0.78	12.1

*Memory side cache capacity, [‡]32MB L2 + 2 $\times$ 16MB L0, [‡]32MB L2 + 4 $\times$ (4MB L1 + 8Mb IMC), [§]192MB + 32MB Frame Buffer, [†]BF16 scaled to 4 INT8 ops

Figure 10.6.6: Comparison tables to hybrid bonded chips and manycore DNN processors.